Methodological Extensions for Probabilistic Earthquake-Triggering Graphs

1 Evaluation of the Current Project and Directions for Completion

1.1 Overall judgment

The project is suitable for an undergraduate thesis, but in its current form it is not yet ready as a publishable paper. The core idea is valuable: earthquake sequences are represented as directed weighted probabilistic triggering graphs, where edges encode the probability that one earthquake triggered another. The student's document already contains a working pipeline from raw files to GraphML and visualization, and it has a clear motivation: moving from deterministic binary declustering toward probabilistic event relations.

For publication, however, the work must be reframed. It should not yet be presented as a new knowledge-graph method. At the moment, a stronger and more defensible framing is:

> *A comparative study of probabilistic directed earthquake-triggering graphs obtained from ETAS-based stochastic declustering and MISD-like data-driven declustering.*

This framing is clearer, more modest, and easier to defend scientifically.

1.2 Important problems in the current formulation

The first important issue concerns the data source description. The NHESS catalog cited by the student is not a general Japan earthquake catalog. It is an updated homogeneous and declustered catalog for South Korea and neighboring regions [1]. The geographical window of that catalog covers only part of Japan and nearby regions. Therefore, before the work can be considered for publication, the student must verify whether the filtered catalog really contains the Japanese events claimed in the analysis. This is especially important if the study intends to include the 2011 Tohoku earthquake sequence.

The second issue is methodological. The document describes MISD as model-independent, but the current implementation uses Gaussian temporal and spatial kernels. Therefore, the current version is better described as a simplified parametric or semi-parametric stochastic declustering algorithm, not as a full implementation of the model-independent stochastic declustering method of Marsan and Lengliné [11].

The third issue is terminology. The word "causal" should be used carefully. The edges in the graph should not be interpreted as proven causal links in the experimental or formal causal-inference sense. They are probabilistic triggering links under a branching process model. More appropriate terms are:

- triggering probability;
- parent–offspring probability;
- branching relation;
- probabilistic triggering link;
- model-dependent causal interpretation.

The fourth issue is that the current reported results are mostly expected results rather than demonstrated results. Statements such as "60–80% of events should be background" or "top events should be $M \geq 6.5$ mainshocks" are hypotheses. In the final thesis, these should be replaced by actual numerical results, tables, plots, sensitivity checks, and case studies.

1.3 Novelty of the idea

The general idea is not completely new. ETAS-based stochastic reconstruction already assigns probabilities that an event is background or triggered by previous events. Zhuang, Ogata, and Vere-Jones [13] introduced this direction for analysing earthquake clustering features through stochastic reconstruction. Similarly, MISD and cascading earthquake-triggering structures are already established; Marsan and Lengliné [11] is one of the standard references for cascading earthquake triggering.

Graph and tree representations of earthquake clusters are also present in the literature. For example, stochastic and nearest-neighbor declustering methods have been compared through tree graph representations and network-analysis tools [14]. More recent studies have also used graph-theoretic quantities such as outdegree centralization, closeness centralization, and average leaf depth in seismic-clustering analysis [6].

Therefore, the novelty should not be stated as:

> *We introduce earthquake graphs.*

A more realistic novelty claim would be:

> *We compare ETAS-based and MISD-like probabilistic triggering graphs on the same earthquake catalog, quantify their agreement and disagreement at event, edge, cluster, and topological levels, and evaluate whether graph-derived features help characterize Type A and Type B strong-aftershock sequences.*

This is a much stronger and more publishable claim.

1.4 Usefulness of comparing ETAS and MISD graphs

Comparing ETAS and MISD-like graphs is useful and should probably become the central direction of the thesis. ETAS and MISD answer related but different questions. ETAS is a parametric statistical-seismology model based on assumptions such as Omori–Utsu temporal decay and a spatial triggering kernel. It is widely used for stochastic declustering and short-term earthquake forecasting. MISD, in contrast, is more data-driven and attempts to infer triggering structure with fewer fixed functional assumptions.

The comparison is useful because it can show which inferred structures are robust and which structures are artifacts of modelling assumptions. However, the comparison must be performed on the same vertex set:

- same catalog;
- same time window;
- same magnitude threshold;
- same coordinate system;
- same magnitude scale;
- same completeness assumptions.

Without this control, differences between the graphs may simply reflect differences in data preprocessing rather than differences between ETAS and MISD.

2 Recommended Graph Comparisons

The comparison should be organized at four levels: event level, edge level, cluster level, and topological level.

2.1 Event-level comparison

For each earthquake j, compare the background probabilities obtained from the two methods:

$$P_{\mathrm{ETAS}}(j \text{ is background}) \quad \text{versus} \quad P_{\mathrm{MISD}}(j \text{ is background}).$$

The following analyses should be included:

- scatterplot of ETAS versus MISD background probabilities;
- Pearson correlation;
- Spearman rank correlation;
- mean absolute difference;
- Bland–Altman-style comparison plot.

The ranking of events should also be compared using weighted out-strength:

$$s_{\mathrm{out}}(i) = \sum_j w_{ij}.$$

This is preferable to simple out-degree because out-degree depends strongly on the chosen edge threshold. Weighted out-strength has a clearer interpretation: it is the expected number of directly triggered offspring.

2.2 Edge-level comparison

For every possible parent–child pair $i \to j$, compare the two edge weights:

$$w_{ij}^{\text{ETAS}} \quad \text{and} \quad w_{ij}^{\text{MISD}}.$$

Useful edge-level comparison measures include:

Question	**Measure**
Do both methods choose the same strongest parent?	Top-1 parent agreement
Do they agree among several likely parents?	Top-k parent overlap
Are the same edges selected after thresholding?	Jaccard index or overlap coefficient
Are edge weights similar?	Weighted correlation, RMSE, cosine similarity
Are parent-probability distributions similar for each child?	Jensen–Shannon divergence or Kullback–Leibler divergence

The threshold $w_{ij} \geq 0.05$ should not be treated as fixed. The student should test several thresholds, for example:

$$\tau \in \{0.01, 0.03, 0.05, 0.10\}.$$

Alternatively, graphs can be compared at equal edge density.

2.3 Cluster-level comparison

After extracting weakly connected components or earthquake cascades, the cluster assignments from ETAS and MISD should be compared using:

- Adjusted Rand Index;
- Normalized Mutual Information;
- Variation of Information;
- cluster size distributions;
- number of singleton/background events;
- number of multi-event clusters;
- largest cluster size;
- proportion of events classified as background or triggered.

This directly addresses the question of whether ETAS and MISD produce similar or different earthquake families.

2.4 Topological comparison

For each cluster graph, compute:

- cluster size;
- depth or maximum generation;
- average leaf depth;
- number and proportion of leaves;
- maximum out-degree;
- weighted out-strength;
- outdegree centralization;
- closeness centralization;
- branching ratio;
- burst-like versus chain-like structure;
- whether the first event, largest event, and most central event are the same.

These measures are useful and align with existing graph-based seismic-clustering literature [14, 6]. However, for publication, the topological analysis should go beyond these standard measures.

3 Statistical Analysis That Should Be Added

The project needs a stronger statistical component. Otherwise, it risks becoming mainly a visualization project.

3.1 Catalog completeness analysis

The student should estimate or justify the completeness magnitude M_c. If the analysis uses only earthquakes with

$$M \geq 4.5,$$

then this threshold must be justified. If smaller events are excluded, the thesis should discuss which triggering chains may be lost.

3.2 Sensitivity analysis

The graph construction should be repeated under several choices:

- magnitude threshold;
- edge threshold;
- temporal bandwidth;
- spatial bandwidth;

- Gaussian versus Omori–Utsu temporal kernel;
- catalog source;
- with and without the 2000-event cutoff.

This is essential for showing that the conclusions are not artifacts of one arbitrary parameter setting.

3.3 Uncertainty analysis

Instead of building only one thresholded graph, the student should sample multiple branching trees from the probability matrix

$$W = (w_{ij}).$$

Then the distributions of graph measures can be computed. This is important because stochastic reconstruction gives probabilities, not a single certain graph.

3.4 Permutation and null-model tests

The student should construct null models by shuffling times, locations, or magnitudes under controlled schemes and then recomputing graph metrics. This would show whether the observed cascade structure is stronger than expected under a random or partially randomized catalog.

3.5 Distributional comparisons

Graph metrics from ETAS and MISD can be compared using:

- Kolmogorov–Smirnov tests;
- Mann–Whitney tests;
- permutation tests;
- bootstrap confidence intervals.

3.6 Case-study validation

The thesis should include a few known Japanese earthquake sequences. The 2011 Tohoku sequence would be an obvious candidate if the catalog actually contains it. The 2024 Noto sequence could also be interesting, but only if the catalog covers it and the data quality is adequate.

For each case study, the inferred family tree should be compared with known sequence behaviour.

4 Methodological Corrections

4.1 Spatial distance

Spatial distance should not be computed naively in degrees. For Japan, spatial distance should be computed in kilometers, preferably using haversine distance or a suitable map projection.

If earthquake depth is included, then a three-dimensional distance may be used. If depth is excluded, this decision should be explicitly justified.

4.2 Temporal kernel

The Gaussian temporal kernel is weak from a seismological point of view. Earthquake aftershock decay is classically closer to Omori–Utsu power-law behaviour, and ETAS explicitly uses this type of temporal decay. A Gaussian kernel may underestimate long-range temporal aftershock links and impose an artificial characteristic time scale.

If the Gaussian kernel is kept, it should be presented as a simplified baseline rather than as the final scientific model.

4.3 Direct and indirect edges

The current rule for classifying direct and indirect edges needs care. If an edge $i \to j$ exists and there is also a path

$$i \to k \to j,$$

this does not automatically mean that the direct edge $i \to j$ is false or should be called indirect. In a probabilistic branching model, direct parent probability and cascade reachability are different concepts.

A better approach is to report:

- direct parent probabilities;
- transitive or cascade reachability;
- expected number of descendants;
- probability that one event belongs to the descendant cascade of another.

4.4 ETAS implementation

The ETAS implementation should be a real baseline, not only a conceptual comparison. There are established tools for fitting space–time ETAS models and performing stochastic declustering, including the R package `ETAS` [8]. Using or validating against an existing implementation would make the work much stronger and more reproducible.

5 Quadratic Complexity Limitation

The quadratic complexity is a real limitation. In the exact formulation, for each event j, all previous events $i < j$ are considered as possible parents. This gives approximately

$$\frac{N(N-1)}{2}$$

candidate parent–child pairs.

For $N = 2{,}000$, this is manageable. For $N = 100{,}000$, it becomes approximately

$$5 \times 10^9$$

candidate pairs. This is not feasible in a standard undergraduate thesis pipeline.

Recent scalable ETAS literature also identifies this as a key limitation. Standard likelihood-based or latent-branching inference for ETAS can scale approximately quadratically, while newer simulation-based approaches aim to make inference feasible for much larger catalogs [9].

5.1 Practical ways to reduce complexity

The following approaches can be used:

Approach	**Usefulness**
Time-window pruning	Only consider candidate parents within a maximum temporal window.
Spatial pruning	Only consider candidate parents within a maximum distance.
Magnitude-dependent windows	Allow larger events to have larger temporal and spatial candidate windows.
KD-tree or Ball-tree search	Efficiently retrieve spatially nearby candidate events.
Sparse matrices	Store only plausible pairs (i, j), not all N^2 pairs.
Top-k candidate parents	Keep only the strongest candidate parents before normalization.
Block processing	Process time-ordered chunks with overlap.
Numba, C++, or parallelization	Useful engineering improvement, but not conceptually sufficient.
Existing ETAS software	Stronger for correctness, validation, and reproducibility.

The best practical solution is:

> *Use a physically justified candidate-parent window first, compute probabilities only for those candidate pairs, store them in a sparse matrix, and perform sensitivity analysis to show that the results are stable as the candidate window expands.*

This is much more defensible than an arbitrary 2000-event cutoff.

6 How This Can Become a Finished Undergraduate Thesis

The final undergraduate thesis can be organized around the following research questions:

1. Can Japanese earthquake catalogs be represented as directed weighted probabilistic triggering graphs?
2. How different are ETAS-based and MISD-like graph reconstructions?
3. Which events are robustly identified as strong triggering hubs?
4. Which graph features distinguish Type A and Type B clusters?
5. How sensitive are the graphs to catalog choice, magnitude threshold, edge threshold, and kernel assumptions?

A coherent thesis structure would be:

1. earthquake catalogs and preprocessing;
2. magnitude homogenization and completeness;
3. ETAS stochastic declustering;
4. MISD or data-driven stochastic declustering;
5. construction of directed weighted triggering graphs;
6. graph comparison methodology;
7. results: event-level, edge-level, cluster-level, and topological comparison;
8. case studies of major Japanese sequences;
9. sensitivity analysis;
10. limitations and future work.

7 Publication Possibility

A publication is possible, but only with a modest and precise novelty claim. The work is unlikely to be publishable as a completely new graph method for earthquakes, because graph and tree representations, ETAS stochastic declustering, MISD, and network-based cluster analysis already exist.

However, it could be publishable as an applied comparative study if it provides:

- a careful Japan-focused analysis;
- robust graph-comparison metrics;
- uncertainty-aware reconstruction;
- a link to Type A and Type B strong-aftershock classification;
- clear sensitivity analysis;
- comparison between ETAS and MISD-like reconstructions.

The closest recent paper is the NESTORE application to Japan by Gentili et al. [16]. That paper applies a machine-learning approach to forecasting strong subsequent earthquakes in Japan. Therefore, the student's paper must clearly distinguish itself.

A possible distinction is:

> *Gentili et al. use ETAS/NESTORE for strong-aftershock forecasting. This project studies and compares the full probabilistic triggering graph structure produced by ETAS and MISD-like reconstruction, and evaluates whether graph topology provides interpretable features for cluster characterization.*

This is a plausible publishable angle.

8 Final Recommendation

For the undergraduate thesis, the project should continue, but the methodology should be narrowed and strengthened. For publication, the contribution should not be oversold. The paper should be framed as a careful methodological and applied comparison, not as a completely new earthquake-graph algorithm.

The must-add elements are:

1. correct catalog definition and geographical coverage;
2. a true ETAS baseline;
3. honest description of the simplified MISD-like implementation;
4. graph comparison metrics;
5. sensitivity analysis;
6. uncertainty analysis by sampling graphs from probabilities;
7. case studies of known Japanese earthquake sequences;
8. clear statement that inferred links are model-dependent triggering probabilities, not proven physical causality.

With these changes, the project can become a strong undergraduate thesis and may be developed into a small but serious applied/statistical seismology paper.

9 Methodological Limitations and Possible Extensions

The current implementation identifies two important methodological limitations: (i) the use of Gaussian temporal and spatial kernels, and (ii) the quadratic complexity of the pairwise parent–child probability calculation. Both limitations can be addressed, but they require different solutions. The Gaussian kernel limitation is primarily a modelling issue, while the quadratic complexity is both a statistical and computational issue.

9.1 Replacing the Gaussian triggering kernel

In the current implementation, the temporal triggering kernel is Gaussian,

$$g_t(\Delta t) = \frac{1}{\sigma_t\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{\Delta t}{\sigma_t}\right)^2\right),$$

and the spatial kernel is also Gaussian. This choice is computationally convenient, but it is not the most natural model for earthquake triggering. Aftershock activity is classically described by Omori–Utsu-type temporal decay, where the triggering effect decreases approximately as a power law rather than as a Gaussian function.

Therefore, the Gaussian version should be treated as a baseline implementation, not as the final scientific model. A stronger methodological structure would be:

Model version	**Description**	**Role in the thesis/paper**
Gaussian MISD-like kernel	Uses Gaussian temporal and spatial kernels.	Simple computational baseline.
Omori–Utsu temporal kernel	Uses a power-law temporal decay closer to classical aftershock models.	Recommended improved version for the undergraduate thesis.
Semi-parametric ETAS kernel	Uses an ETAS-type triggering function with estimated parameters.	Reference statistical-seismology model.
Nonparametric / log-binned MISD kernel	Learns the triggering density directly from weighted parent–child pairs.	Best direction for a publishable extension.

A more seismologically meaningful replacement for the Gaussian temporal kernel is

$$g_t(\Delta t) = \frac{1}{(\Delta t + c)^p},$$

where $c > 0$ avoids singularity near zero and $p > 0$ controls the temporal decay. A corresponding spatial kernel may be written as

$$g_s(\Delta r) = \frac{1}{(\Delta r + d)^q},$$

where $d > 0$ and $q > 0$ control the spatial decay.

A possible semi-parametric triggering kernel is therefore

$$K_{ij} = \rho(m_i) \cdot (\Delta t_{ij} + c)^{-p} \cdot (\Delta r_{ij} + d)^{-q},$$

where $\rho(m_i)$ is the productivity term associated with the magnitude of the potential parent event i.

This version is still not fully model-independent, but it is a clear improvement over the Gaussian kernel because it reflects the long-tailed temporal behaviour of aftershock activity.

9.2 Towards a more MISD-like nonparametric kernel

For a more model-independent version, one can estimate the triggering density from the data. Instead of assuming a fixed Gaussian or power-law form, the algorithm can estimate a kernel over binned values of

$$\log_{10}(\Delta t), \qquad \log_{10}(\Delta r), \qquad m_i.$$

During the expectation step, the current parent probabilities w_{ij} are used as weights. During the maximization step, the triggering density is updated from the weighted empirical distribution of parent–child pairs.

In simplified form, the update can be written as

$$\widehat{K}(B_t, B_r, B_m) = \frac{\sum_{i<j} w_{ij} \mathbf{1}\{\Delta t_{ij} \in B_t,\, \Delta r_{ij} \in B_r,\, m_i \in B_m\}}{\sum_{i<j} \mathbf{1}\{\Delta t_{ij} \in B_t,\, \Delta r_{ij} \in B_r,\, m_i \in B_m\}},$$

where B_t, B_r, and B_m are bins for temporal distance, spatial distance, and parent magnitude.

This is closer to the spirit of model-independent stochastic declustering, because the shape of the triggering kernel is learned from the data rather than imposed in advance.

For the undergraduate thesis, implementing the Omori–Utsu temporal kernel would already be a strong improvement. For a paper, a nonparametric or log-binned MISD-like kernel would make the contribution more original.

9.3 Reducing the quadratic complexity

The current algorithm has quadratic complexity because, for every earthquake j, all previous earthquakes $i < j$ are considered as possible parents. This gives approximately

$$\frac{N(N-1)}{2}$$

candidate parent–child pairs.

For $N = 2{,}000$, this is manageable. For $N = 100{,}000$, it becomes approximately 5×10^9 possible pairs, which is not feasible in a standard undergraduate project pipeline.

The correct solution is not to keep only the first 2000 events. That may remove important seismic structure. A better solution is to keep the full catalog but restrict the set of candidate parent events.

For each event j, define the candidate parent set

$$\mathcal{C}(j) = \{i < j : 0 < t_j - t_i \le T_{\max}(m_i), \quad r_{ij} \le R_{\max}(m_i)\}.$$

Here $T_{\max}(m_i)$ and $R_{\max}(m_i)$ may depend on the magnitude of the potential parent earthquake. Larger events should be allowed to have longer temporal and spatial influence windows.

Then the probabilities are computed only for

$$i \in \mathcal{C}(j),$$

not for all $i < j$. This changes the practical complexity from

$$O(N^2)$$

to approximately

$$O(Nk),$$

where k is the average number of candidate parents per event.

9.4 Practical sparse implementation

The recommended implementation is:

1. Sort all earthquakes by time.
2. For each event j, search only previous events within a maximum temporal window.
3. Use a KD-tree or Ball-tree to identify spatially nearby candidate parents.
4. Compute w_{ij} only for the candidate parent set $\mathcal{C}(j)$.
5. Store the probabilities as a sparse edge list:

$$(i, j, w_{ij}).$$

6. Normalize over the candidate parents and the background term:

$$w_{0j} + \sum_{i \in \mathcal{C}(j)} w_{ij} = 1.$$

This avoids constructing a dense $N \times N$ matrix and makes it possible to use much larger catalogs.

The pruning windows should not be arbitrary. The thesis should include a sensitivity analysis over several choices, for example:

$$T_{\max} \in \{30, 90, 180, 365\} \text{ days}$$

and

$$R_{\max} \in \{50, 100, 200, 400\} \text{ km}.$$

If the main conclusions are stable across these choices, the approximation becomes scientifically defensible.

9.5 Topological analysis

Basic graph measures should be included, but they are not sufficient for novelty. Several graph-based and tree-based analyses of earthquake declustering already exist in the literature. Therefore, the topological part of the thesis should go beyond simple degree, centrality, and component-size measures.

9.5.1 Standard graph measures

The following measures should be included as a necessary baseline:

$$s_{\text{out}}(i) = \sum_j w_{ij},$$

where $s_{\text{out}}(i)$ is the weighted out-strength of event i. This quantity has a direct interpretation: it is the expected number of direct triggered offspring.

Other useful standard quantities are:

- weighted in-strength;
- number of weakly connected components;
- largest component size;
- cluster depth;
- maximum generation;
- number and proportion of leaves;
- branching ratio;
- out-degree centralization;
- closeness centralization.

These measures are useful, but they should be treated as baseline descriptive tools.

9.5.2 Cascade morphology measures

A more interesting contribution is to describe the morphology of earthquake cascades. Different clusters may have very different shapes:

- star-like clusters, where one main event triggers many direct aftershocks;
- chain-like clusters, where triggering propagates across many generations;
- bushy clusters, where several secondary events trigger their own aftershocks;
- mixed clusters, where a mainshock is followed by several strong branches.

To quantify these structures, the following tree-shape measures can be used:

Measure	**Interpretation**
Horton–Strahler order	Measures hierarchical branching complexity.
Sackin index	Measures total depth of all nodes; large values indicate deeper cascades.
Colless index	Measures tree imbalance; useful for distinguishing symmetric from asymmetric cascades.
Leaf-depth distribution	Shows whether aftershocks are concentrated near the main event or spread across several generations.
Subtree-size distribution	Identifies whether the sequence contains several important subfamilies.
Generation entropy	Measures whether events are concentrated in one generation or distributed across many generations.

These measures could be especially useful for comparing Type A and Type B earthquake clusters. Type A clusters, which contain strong subsequent events, may have deeper, more branched, or more imbalanced cascade structures than Type B clusters.

9.5.3 Persistent threshold topology

The strongest potential novelty is to avoid analysing only one thresholded graph. Instead of fixing a single threshold such as $w_{ij} \geq 0.05$, define a family of graphs

$$G_\tau = (V, E_\tau),$$

where

$$E_\tau = \{i \to j : w_{ij} \geq \tau\}.$$

Then vary the threshold:

$$\tau \in \{0.50, 0.40, 0.30, 0.20, 0.10, 0.05, 0.03, 0.01\}.$$

This produces a filtration of graphs. As τ decreases, more edges appear and the graph becomes denser. The analysis can then focus on which structures persist across many thresholds.

Useful persistence-based questions are:

- Which clusters appear at high threshold and remain stable as the threshold decreases?
- Which triggering hubs remain important across many thresholds?
- Do ETAS and MISD identify the same persistent clusters?
- Are Type A clusters more persistent or more topologically complex than Type B clusters?
- Are some edges method-dependent, appearing only under ETAS or only under MISD?

This is more robust than analysing a single arbitrary threshold.

9.6 Graph ensembles instead of a single graph

Because w_{ij} are probabilities, the most statistically honest approach is to analyse an ensemble of possible triggering forests.

For each event j, sample its parent from

$$\{0, 1, 2, \ldots, j-1\}$$

with probabilities

$$w_{0j}, w_{1j}, w_{2j}, \ldots, w_{j-1,j},$$

where 0 denotes a background event.

Repeating this procedure B times gives B possible triggering forests:

$$G^{(1)}, G^{(2)}, \ldots, G^{(B)}.$$

Then each graph measure can be reported with uncertainty:

$$\widehat{\theta} = \frac{1}{B} \sum_{b=1}^{B} \theta(G^{(b)}),$$

together with empirical confidence or credible intervals.

This would allow the student to report uncertainty for:

- number of clusters;
- largest cluster size;
- cascade depth;
- branching ratio;
- hub ranking;
- Type A/Type B classification;
- tree imbalance;
- persistence measures.

This is a much stronger approach than building a single graph after thresholding.

9.7 Recommended final direction

For the undergraduate thesis, the following improvements are sufficient and realistic:

1. Replace the Gaussian temporal kernel with an Omori–Utsu-type temporal kernel, or at least compare both.
2. Compute spatial distance in kilometers, not in degrees.
3. Replace the fixed 2000-event cutoff with candidate-parent pruning.
4. Use sparse edge representation.
5. Compare ETAS-based and MISD-like graphs on the same catalog.
6. Add standard graph measures and cascade morphology measures.
7. Perform sensitivity analysis over thresholds, temporal windows, and spatial windows.

For a publishable paper, the stronger contribution would be:

1. implement a nonparametric or log-binned MISD-like kernel;
2. compare it with ETAS stochastic declustering;
3. analyse ensembles of possible triggering forests instead of one fixed graph;
4. introduce persistent threshold topology for probabilistic earthquake-triggering graphs;
5. compare cascade morphology between Type A and Type B clusters;
6. show robustness across catalog choices and modelling assumptions.

A possible paper title would be:

Uncertainty-aware topological comparison of ETAS and model-independent stochastic declustering graphs for Japanese earthquake sequences.

The central novelty should not be stated as the construction of earthquake graphs in general, because graph and tree representations already exist. A stronger and more defensible novelty claim is:

> We compare ETAS-based and MISD-like probabilistic triggering graphs using event-level, edge-level, cluster-level, and persistent topological measures, and we quantify the uncertainty of graph-derived cascade features through ensembles of sampled triggering forests.

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